

SAN DIEGO DEVELOPMENT OPPORTUNITY FINDER

A tract-level screening tool for comparing neighborhood conditions



Presented By
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PROBLEM STATEMENT

Finding development opportunities takes a lot of research.

Rent • Income • Transit • Walkability • Schools • Climate • Safety

Reviewing these factors tract by tract across a county can take a lot of time and resources.



PROJECT GOALS



Original Concept

Create one **developmental opportunity score** for each tract, ranking them based on key features that would flag them as a good place to develop a specific product (schools, housing, transit, retail, etc.)

Facing Reality

Neighborhood factors interact in complicated ways, and it's not easy to build a single score to rank them or identify what products might perform well. I could use the scores as a way to flag a tract that could use more indepth analysis.

Final Approach

Predict expected rent with model

Find tracts where 'actual rents' that were unusually low

Group similar neighborhoods

Use additional filters to narrow the search



WRANGLING THE DATA



Demographics + Housing
2024 Census Data



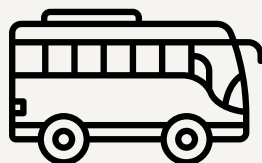
Crime + Safety
City of San Diego



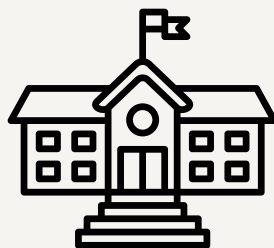
Walkability
EPA



Climate Risk
FEMA



Transit
City of San Diego



Schools
California Public Education Data

STATS

6+

Datasets

727

Tracts

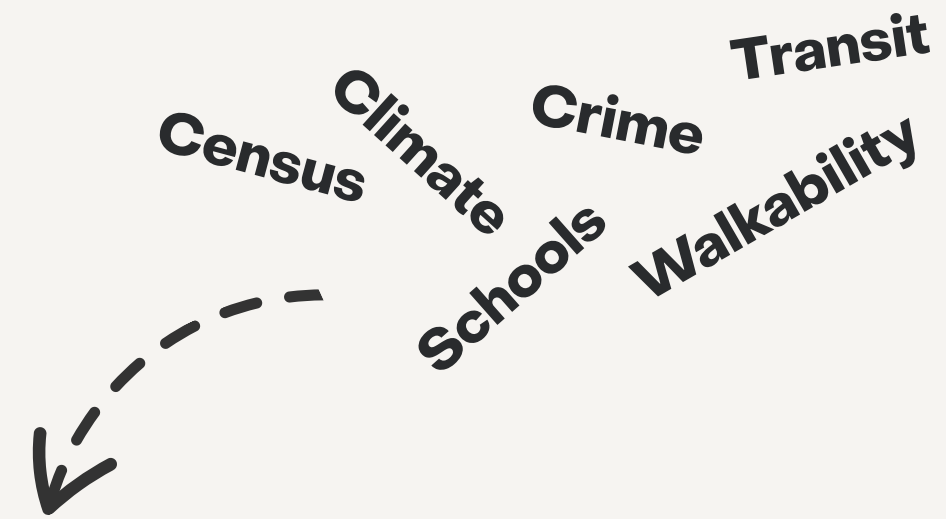
75

Initial Features

20

Final Features

CLEANING THE DATA



Clean *each* dataset separately

- Rename columns
- Fix tract IDs
- Handle missing values
- Create rates/scores

Spatial matching + Aggregation

Match crime incidents, transit stops, schools and districts, and block groups to census tracts

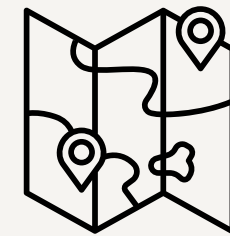
Merge + Clean Again

Merge on census tracts into one master dataset, then clean one more time for null values

CLEANING THE DATA



Major Painpoints



Geography

Different tract years/boundaries
Schools and police data needed
spatial matching



Missing data

199 walkability values (27%)
79 original rent values (10%)
8 income values (1%)



Coverage

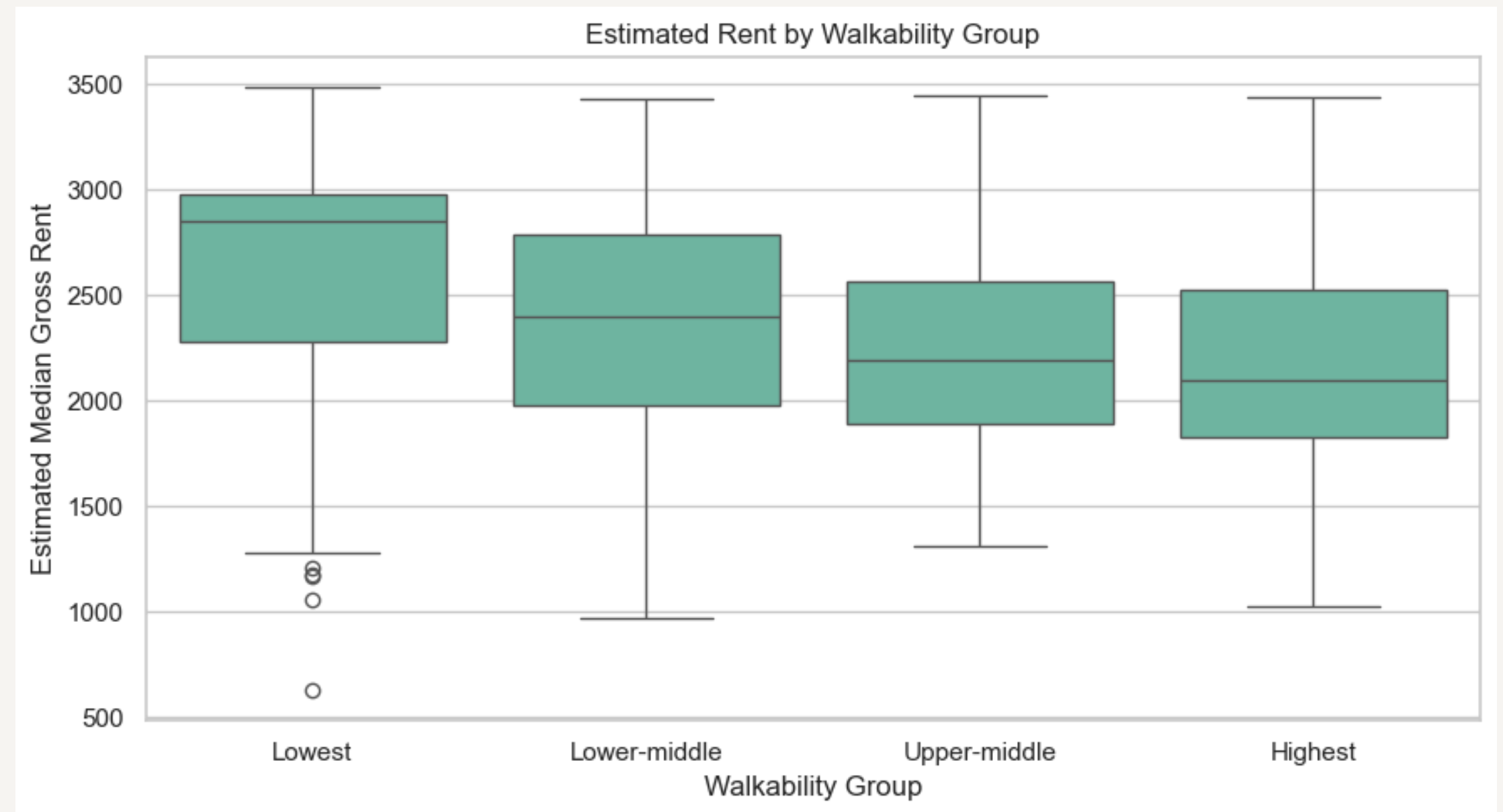
SDPD safety data doesn't cover all of
San Diego County

EXPLORATORY DATA ANALYSIS

Walkability Didn't Automatically Mean Higher Rent

- Lowest-walkability tracts had the highest median rent*
- Rent generally decreased across walkability groups
- Large overlap between groups

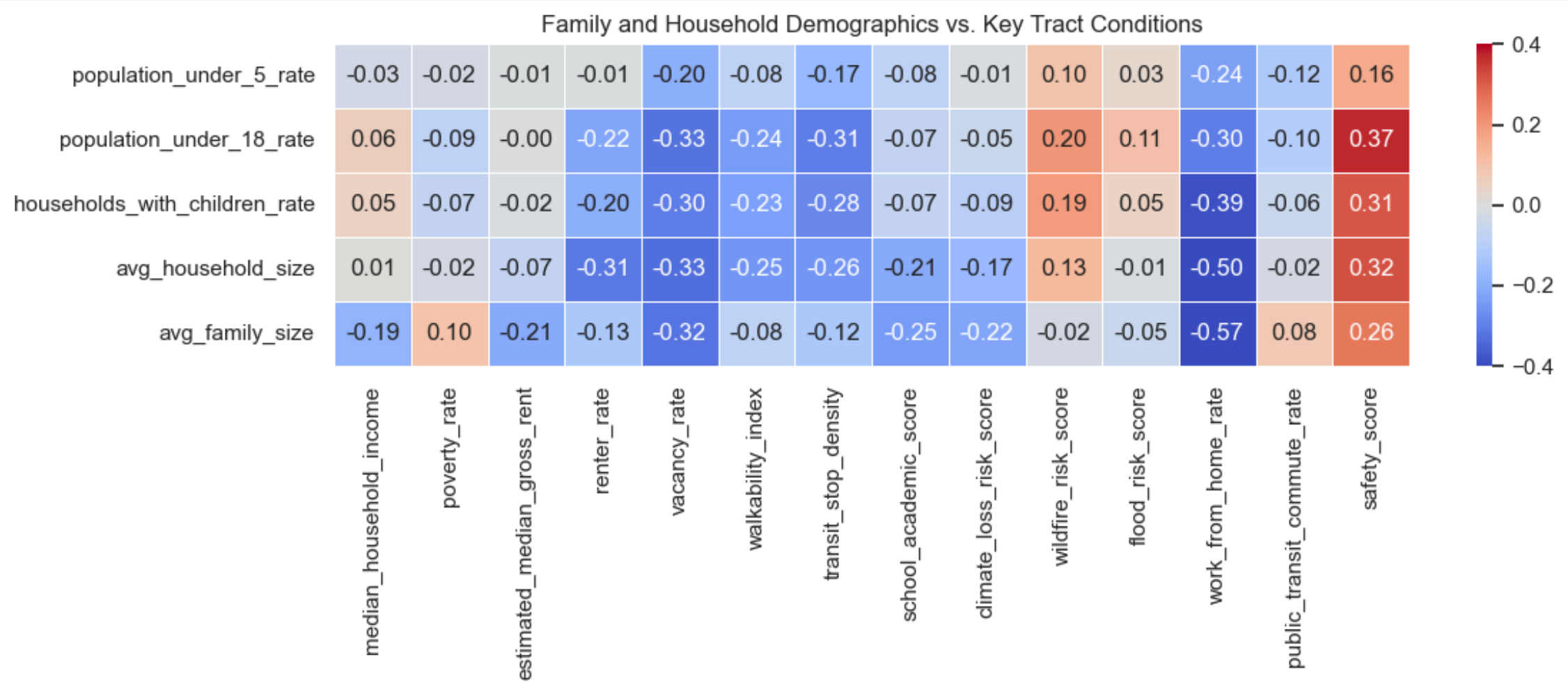
**Median rent is for the entire tract, it didn't account for the size of the rental. It's possible that rent was higher in less walkable areas due to the size of the rental properties. For future analysis I'd inspect bedroom count, housing types, and other factors.*



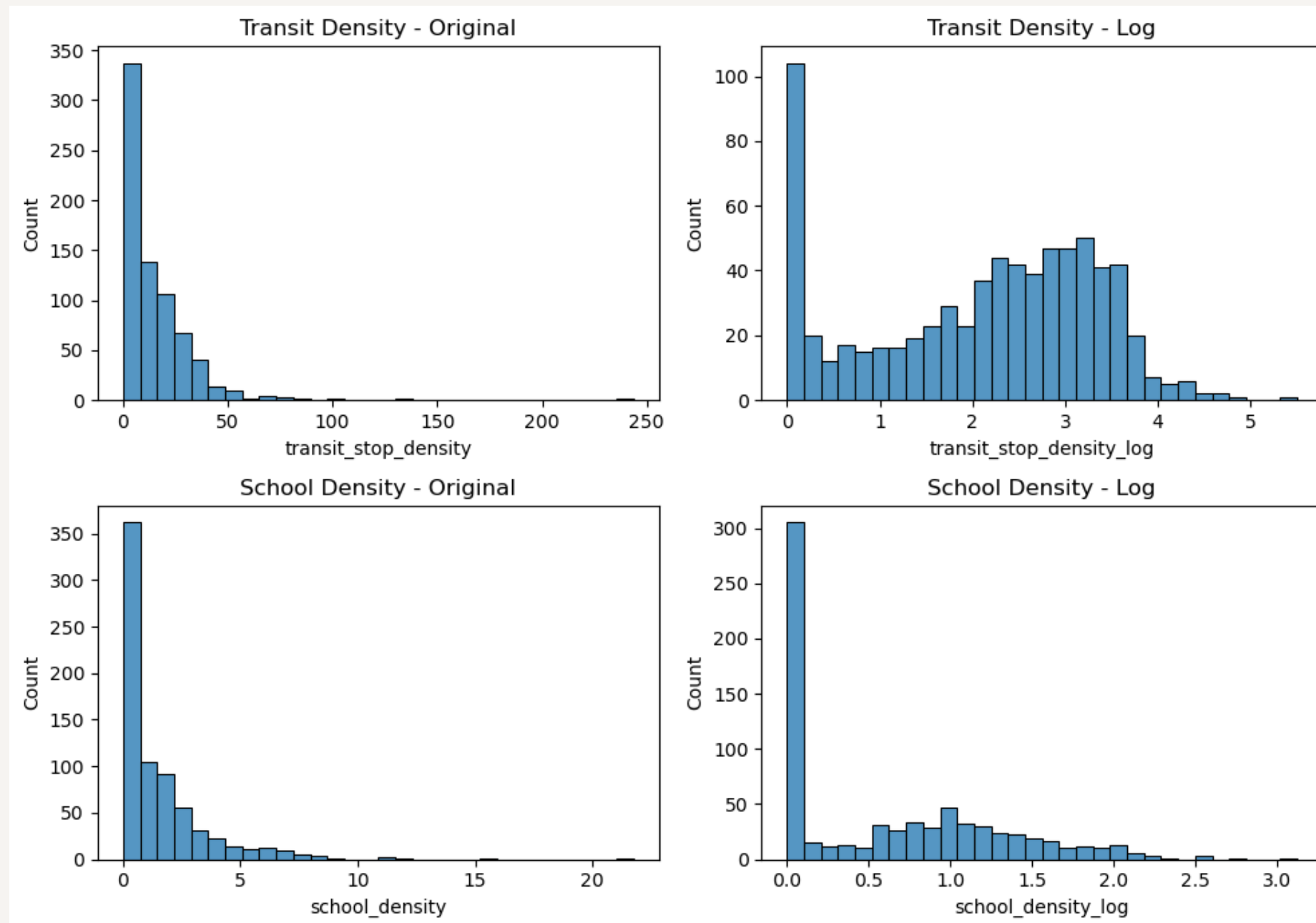
EXPLORATORY DATA ANALYSIS

Different Neighborhood Features Moved Together

- Walkability and transit density were strongly related
- Family-oriented tracts had larger households and different transportation patterns
- Lower-walkability tracts tended to have higher wildfire risk



PREPROCESSING + MODEL SETUP



Preparing 20 Features for Modeling

Transformations

- Transit density → log transformed
- School density → log transformed

Model split

- Training: 581 tracts
- Test: 146 tracts
- 80/20 split

Scaling

- StandardScaler
- Fit on training data only

REGRESSION MODEL COMPARISON

Target Variable: Median Gross Rent

Model	Test R ²	RMSE	MAE
Linear Regression	0.566	\$364	\$278
Tuned Random Forest	0.535	\$377	\$281

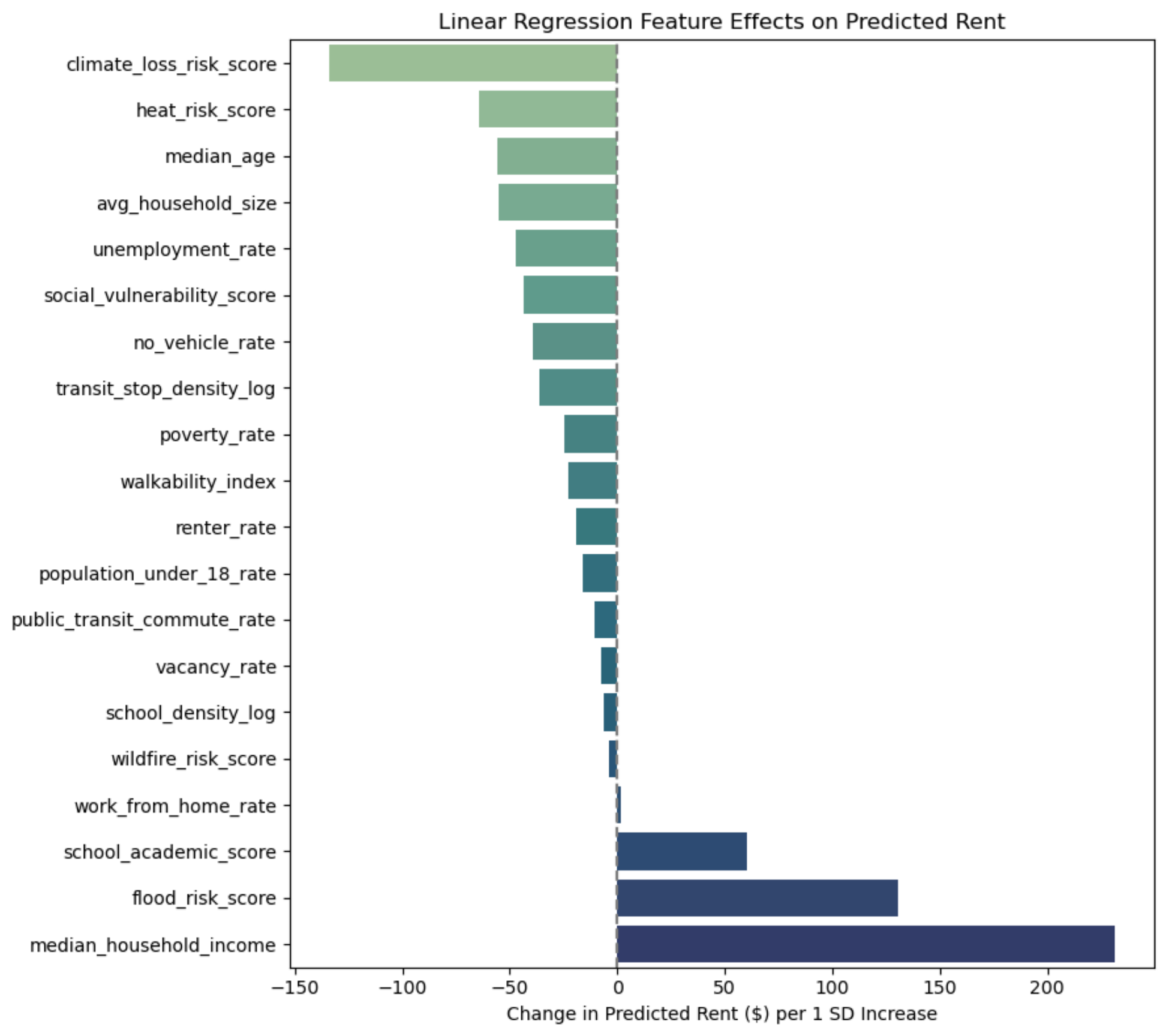
Final model: Linear Regression



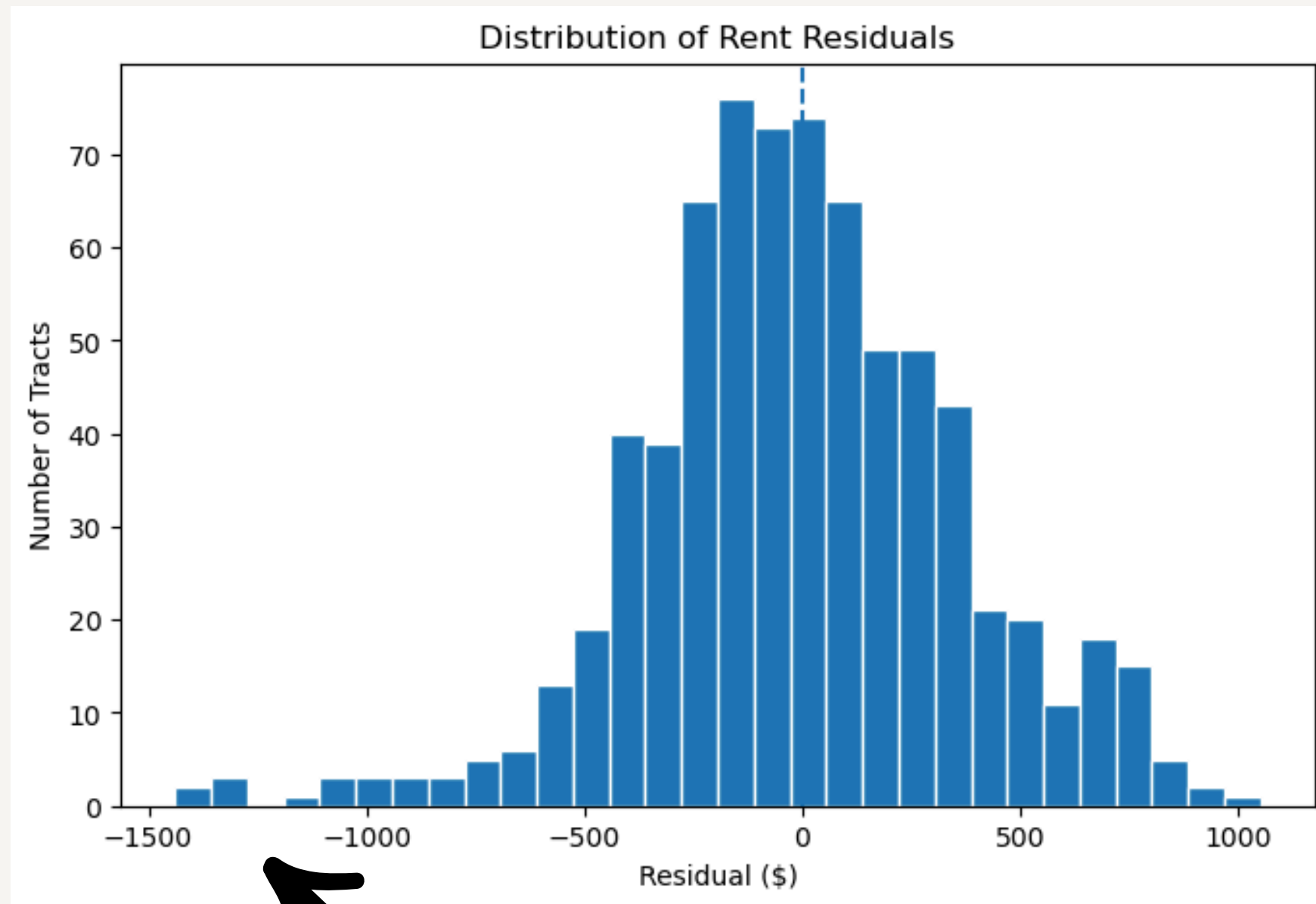
FEATURE IMPORTANCE

Income Had the Strongest Relationship With Predicted Rent

Coefficients show model associations after controlling for the other included features. They are ***not*** cause and effect.



RESIDUAL SCREENING



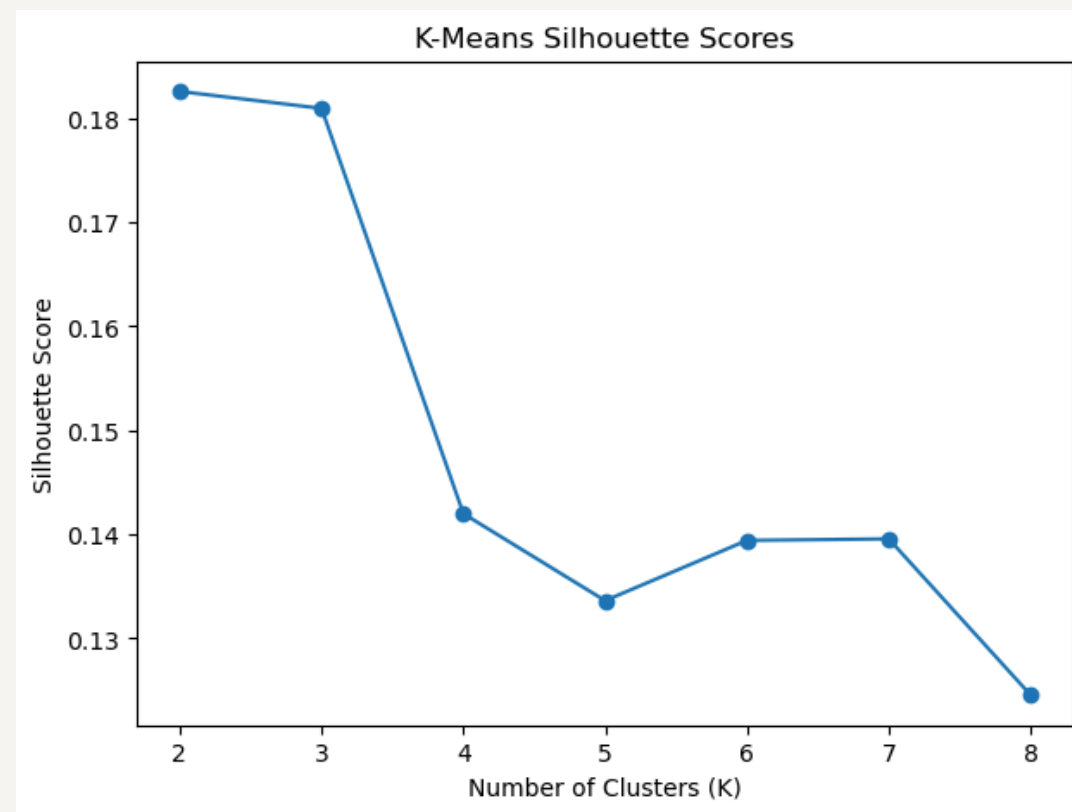
Bottom 10% in question

Where Is Actual Rent Much Lower Than Expected?

- Bottom 10% cutoff $\approx$ -\$415
- 73 of 727 tracts flagged

Residual = Actual Rent - Predicted Rent

UNSUPERVISED LEARNING



K=3 silhouette score ≈ 0.18 , so the groups overlap and should be treated as broad profiles rather than strict categories.

Three Broad Neighborhood Profiles



Affluent Suburban Families

Higher income, larger households, less transit, higher wildfire risk



Transit Family Renters

More renters/children, lower income, higher transit reliance



Urban Renters

Highest walkability/transit access, smaller households, more renters

CLUSTER PROFILES

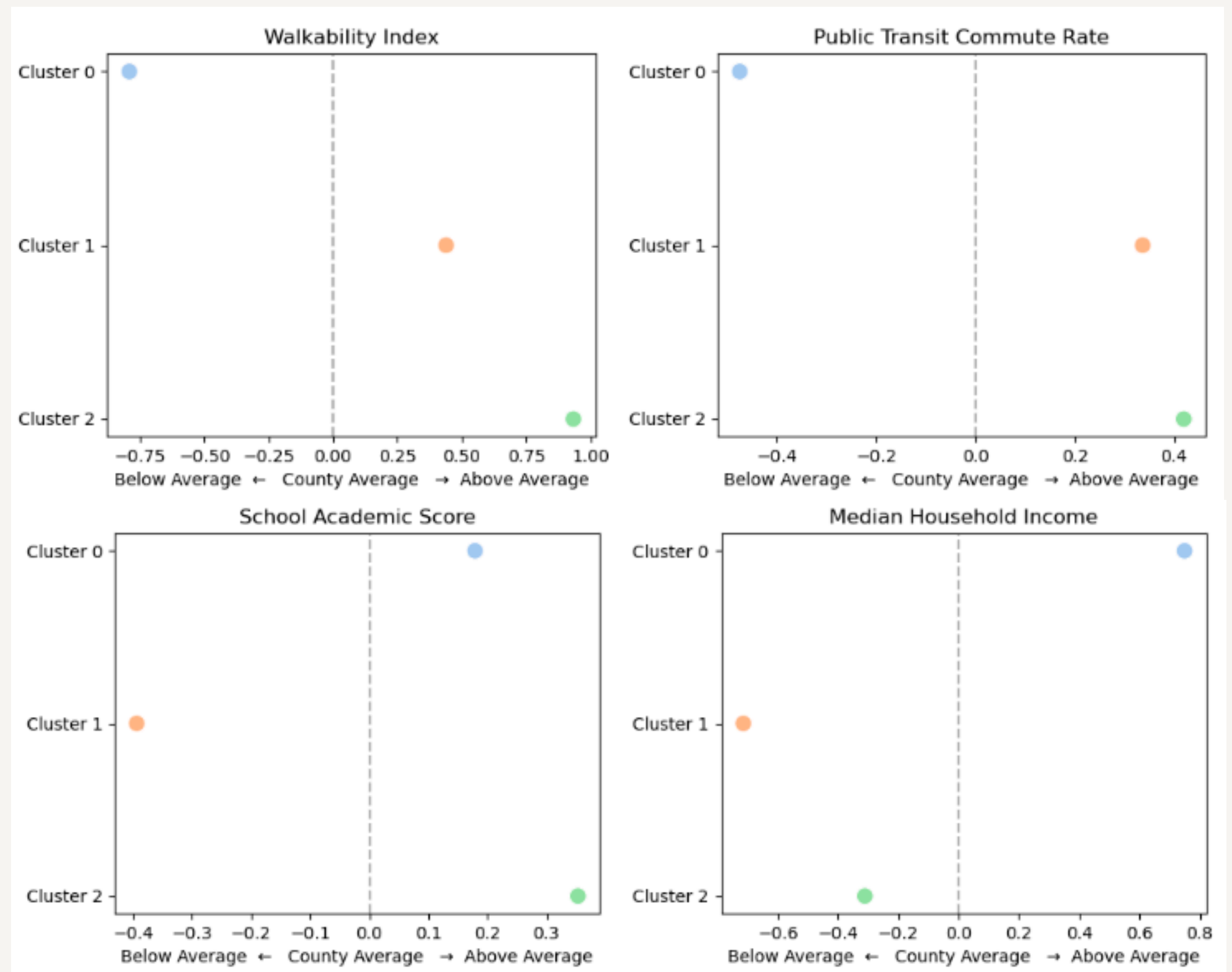
Cluster Profiles Compared with County Averages

Values show whether each cluster is above or below the county average for each feature.

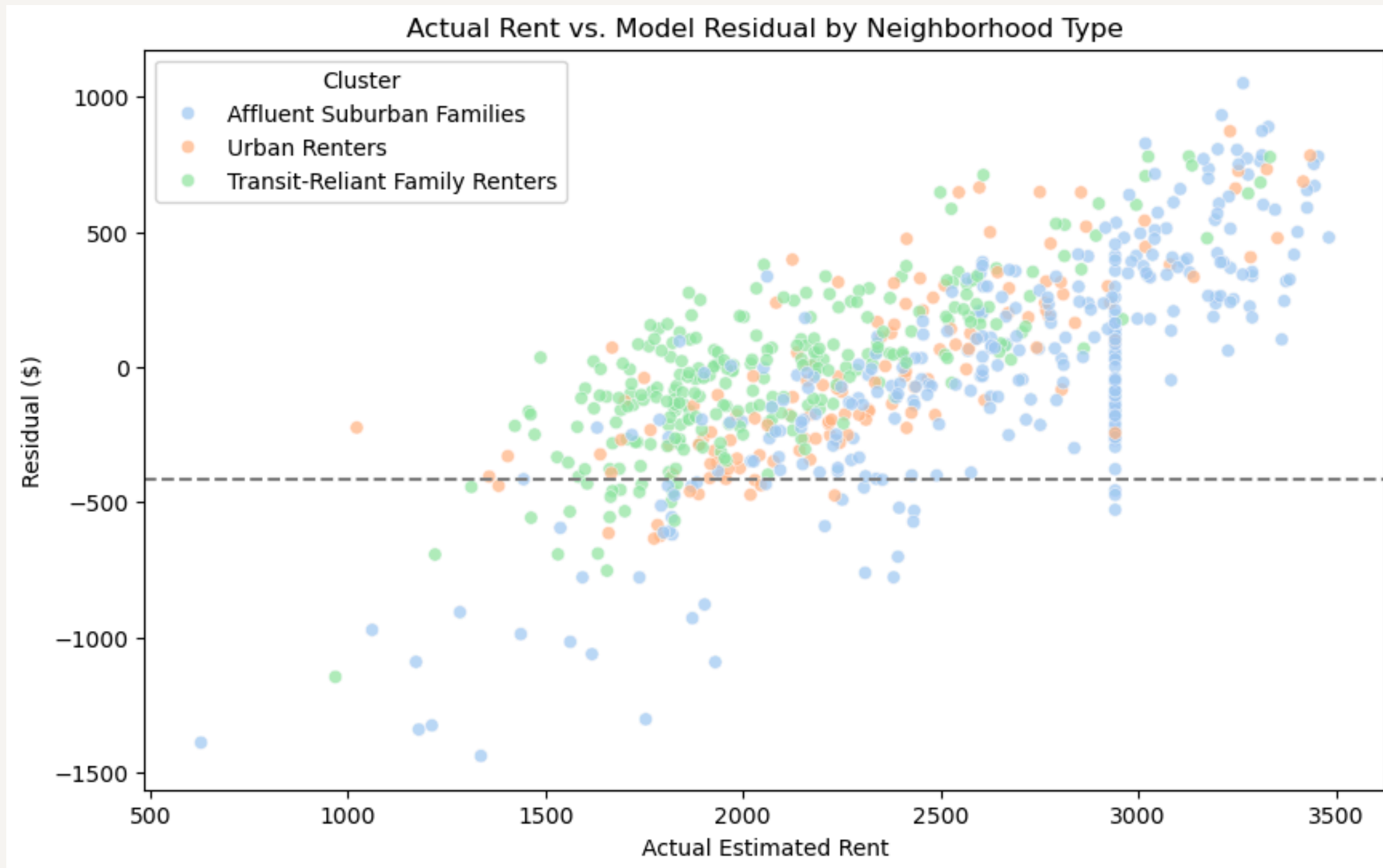
I created subplots for all 20 features so I could clearly see how each cluster related to the county averages and each other.

Key

- Affluent Suburban Families
- Transit Family Renters
- Urban Renters



INSPECTING MODEL RESULTS



Residual Screening Across Neighborhood Types

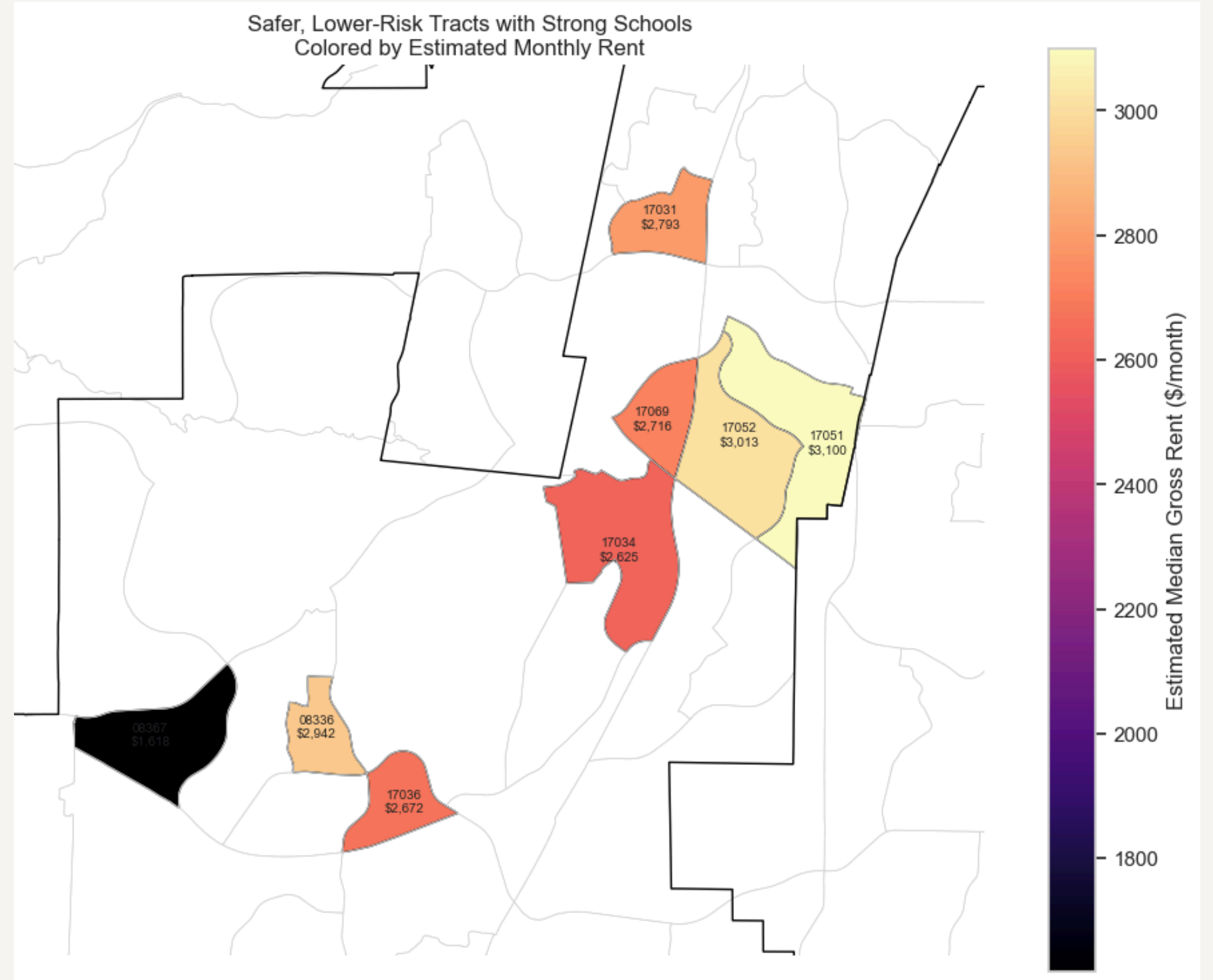
- Lower-than-expected rent appeared in all three cluster types
- Affluent Suburban Family tracts had some of the largest negative residuals
- Cluster type helped describe neighborhoods, but residuals worked better as a separate screening signal

BUILDING A SHORTLIST

From 727 tracks down to 8

727 county tracts
↓
311 inside SDPD jurisdiction
↓
Top 25% safety
↓
Bottom 25% climate-loss risk
↓
Compare schools + rent + location

Each of these tracts happen to be in the same, highly-rated school district. It could be a good opportunity for someone building family housing to look into these tracts.



RECOMMENDATIONS + NEXT STEPS

Recommendations

- Use the dataset as an early site-screening tool with searchable filters
- Use residuals to identify unusual rent relationships
- Compare tracts within similar neighborhood types

Next Steps

- Build a filterable site-selection tool for developers and investors
- Add parcel availability, zoning, market data, housing/unit characteristics, ZIP codes, and school boundaries
- Add countywide safety data from multiple agencies
- Improve school assignment accuracy and explore more SANDAG GIS data
- Test models using different target variables and features



Lesson Learned

The scope became overwhelming early because I was working with a lot of different datasets before fully understanding each one. If I did this again, I'd spend more time cleaning, checking distributions, and exploring each dataset before merging. I'd also reduce the scope and treat each source as a smaller analysis first, then combine only the most useful features into the final model.